$$\frac{\partial \mu}{\partial \mu^0} = \prod_{i=1}^{n} H\big(\phi^i(\mathbf{x})\big) \tag{27}$$

For the shear modulus $\mu^j$ of the $j$-th component, $\frac{\partial \mu}{\partial \mu^j}$ is given by:

$$\frac{\partial \mu}{\partial \mu^j} = 1 - H\big(\phi^j(\mathbf{x})\big) \tag{28}$$

# Results

## *Case 1: A layered rectangular structure*

In the first numerical example, we analyze a layered rectangular structure, illustrated in Fig. 6. The dimensions of this structure are 1.0 mm × 0.2 mm. To discretize the problem domain, we utilize 7200 linear triangular elements. In the forward finite element simulation, a concentrated force is applied to the top edge, while the bottom edge was restricted the vertical motion and the center point is fully fixed. In the inverse problem, our focus lies in utilizing displacement datasets collected from the top edge as inputs for the inverse scheme. Acknowledging that relying on a single surface displacement dataset may not produce optimal reconstruction results, we change the location of the concentrated force and incorporate multiple surface displacement datasets to solve the inverse problem. The target shear modulus distributions for one (the homogenous case), two, and three layers are depicted in Fig. 6(a), (b), and (c), respectively. The shear modulus values of each layer in the structure are of a similar order to that of skin tissue [36]. Considering the uncertainty regarding the total number of layers, we initially assume a configuration with six layers for the inverse problem, as depicted in Fig. 7(a). For the explicit inverse scheme, we utilize 10 control points to interpolate each interface. Thus, the total number of optimization variables is 56 including 50 geometric variables and 6 shear moduli.

From Fig. 7, it can be seen that both the shape of the interfaces between the neighboring layers and the shear modulus values of each layer varies and finally reach to the target shear modulus distribution (Sample 1). Note that it takes only 52 iterations for convergence. Figs. 8–10 are the reconstructed shear modulus distributions for Samples 1–3 using explicit and implicit inverse approaches for the noise free cases, respectively. In the implicit inverse method, the total number of optimization variables is linearly proportional to the mesh size. In this case, the total number of optimization variables is 3691, which is equal to the total number of nodes. The reconstructed results demonstrate that both inverse methods can yield decent reconstruction with noise free cases. Further, the implicit method fails to recover the heterogeneity at the upper corner compared with the results by the explicit inverse method. Since the total number of optimization variables is remarkably reduced in the explicit inverse method, the total number of minimization iterations is reduced. The total number of minimization iterations for the explicit inverse method is approximately 10 times less than that required for the implicit inverse method. Furthermore, the shear modulus reconstruction obtained through the implicit

**[Figure 6 — placeholder]**

Three layered rectangular structures (1.0 mm × 0.2 mm), each showing the target shear modulus (SM) distribution in MPa with a jet-style colorbar on the right.

**(a) Sample 1**: homogeneous, single layer, SM = 3.00 MPa (uniform blue). Downward arrows on top edge indicate concentrated force locations. Red text "displacement measurement" with leftward arrows on left side.

**(b) Sample 2**: two layers. Bottom layer SM ≈ 3 MPa (blue), top layer SM ≈ 10 MPa (red). Green circles on top boundary, red circles on bottom boundary mark displacement measurement points.

**(c) Sample 3**: three layers (blue–red–blue sandwich). SM ranges 1.00–10.00 MPa. Same boundary marker pattern as (b).

All three samples share the same mesh (7200 triangular elements) and boundary conditions (top-edge force, bottom-edge vertical constraint, center point fixed).

**Fig. 6.** The target shear modulus distribution of three samples with different number of layers (Unit: MPa). SM stands for shear modulus.